

Time Allotted : 50 min
Total weight : 20 %

Name _____ IDNo _____ Section _____

General Directions:

- Write your name and ID No. both on this Question paper and answer sheet.
- Do not use pencil and red pen.
- Do not hand any materials related to the course.
- Please read the instructions carefully and attempt all the questions.
- Cheating automatically nullifies your total scores.
- It is strictly forbidden to use mobile phones in the examination. Unless permitted, no other electronic devices and materials are allowed in the examination room.

PART I: Choose the best answer from the given alternatives and give your answer on the space provided in the answer sheet. [1 pt each]

1. Which efficiency metrics define the difference in packet arrival time of information when it arrived at destination.
A. Delivery B. Jitter C. Accuracy D. Timeliness
 2. Which component of internetworking is considered as a secured extension of organizations local area network, so as to access the network Anywhere anytime
A. LAN B. Intranet C. Internet D. Extranet
 3. Identify the IPV6 address that is analogous with Routable IP address of version 4.
A. Global Address C. Mapped Address
B. Link local Address D. Compatible Address
 4. Which component in an IP address Identifies what bits in the IP address are to be used to represent the network/subnet portion of an IP address.
A. Network Component C. Host Component
B. Subnet Mask D. Broadcast Component
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PART II : Answer the following Question accordingly [16 pt]

1. What do Networking models explain in computer networks and explain the most notable network model. [3 pt]
2. Assume a PDU “Hey” prepared at the Application layer. Use Non-Return-to-Zero Inverted line encoding scheme to represent pieces of the PDU separately at the physical layer. [3 pt]
3. A **DataCom company** wants you to **design logical addressing** for its small networks, i.e Admin network, User network, Library network, CEO network, Developers network and Security network. For the purpose of its future scalability you received a network address of 172.16.10.0/24. Determine the following [4pt]
 - A. The subnet mask
 - B. The network address of each sub-network
 - C. The address space of all sub-network
 - D. The broadcast address of subnetwork
4. Jeff Bezos wants to extend the Amazon network to Ethiopia as a separate new Autonomous network with more than 60,000 hosts and got network address of 172.16.10.0. A.A_NET has 31,000 hosts, Hawassa_NET has 16,000 hosts, Jimma_NET has 8000 hosts, Bahirdar_NET has 2000 hosts, Adama_NET has 2000 Hosts. Beside, to connect the nets Jeff also plans to use 2 separate Routers(NORTH_rtr and SOUTH_rtr). NORTH_rtr connects A.A_NET, Adama_NET and connects to SOUTH_rtr, where as SOUTH_rtr connects Jimma_NET, Bahirdar_NET, Hawassa_NET and with NORTH_rtr. Based on the information. [6pt]
 - A. Draw the high level topology
 - B. Determine the Bit-to borrow for each NET
 - C. The subnet mask of each NET
 - D. The network address of each NET
 - E. The address space of each NET
 - F. The broadcast address of subnetwork

Answer Sheet

Result: _____

Name _____ ID NO _____ Sec _____

Part I: Multiple Choice [24pt]

1.	4.	7.	10.	13.	16.	19.	22.
2.	5.	8.	11.	14.	17.	20.	23
3.	6.	9.	12.	15.	18.	21.	24.

Part II: Short Answer [16pt]

1.
